Secure CAN Monitor: Progress Report (2026-11-12)

Secure CAN Monitor: Progress Report (as of 2026-11-12)

Evaluation: Saturday 2026-11-14. Sources: the status board and dated log in `Secure_CAN_Monitor_Plan_8_Components.md`, the component records in this folder, the board logs in `Part_E_Verification/baseline/`, and the Pi checked read-only on 2026-11-12.

1. What the project is

A **detection-only** secure CAN monitor on Raspberry Pi 3 B (TF-A + OP-TEE 4.10 + Linux, MCP2518FD on `can0`). Core 1 runs in Secure World and watches a heartbeat challenge and the CAN-traffic deadline. The Pi 3 has no TZASC, so nothing here prevents anything. Keys are prototype values readable by root Linux. The event log is RAM-only with no hash chain. **All frames used so far are software-injected through the PTA or loopback, not CAN-bus traffic.**

2. State of the Pi (checked 2026-11-12)

8 GB spare card (`SD8GB`, serial 0x6d280b3c); hardened c16384 monitor image (`armstub8.bin` sha256 `775ce384...`); N-04 DTB installed; `verify_boot` 16 pass / 0 FAIL / 2 skipped; monitor `init_ok=1`, `starved=0`; `teeclnt` empty. The 32 GB original card was never written.

3. Components

#	Component	Status	Evidence
C1	Baseline and hardware readiness	**Done** (software gate); hardware items open	`baseline/COMPONENT_1_RECORD.md`; N-04 resolved 2026-11-12
C2	Secure monitor foundation	**Closed**: 51 PASS / 2 PARTIAL / 0 FAIL of 53	`monitor/COMPONENT_2_2_INVENTORY.md`, `COMPONENT_2_GATE_CHECK.md`
C3	Core 1 integration	**Done with accepted deviations**	`monitor/COMPONENT_3_...md`
C4	Timing and measurement	**Closed with limits** for Mode F; floods, sender, histogram not done; rule (b) fails with the plan's provisional numbers	`monitor/COMPONENT_4_...md`, `monitor/derived/derive_params_20261112_plan_values.txt`
C5	Heartbeat and traffic deadline	**Done with limits (accepted)**: board 26 PASS / 0 FAIL / 1 skip, plus 2 real reboots; host tests for T11a/T11b/T10b added	`baseline/c5_board_tests_run2_full_20261112.txt`, `monitor/COMPONENT_5_...md`

#	Component	Status	Evidence
C6	Hardening (M1)	**Partial**: idle caps verified on the board; IDLE_EXPIRED and the UART line written, host-tested, built, **not flashed**	`baseline/c6_negative_idle_caps_20261112.txt`, `monitor/COMPONENT_6_STAGE_AB_EVIDENCE.md`
C7	Independent CAN and cross-checking	**Not started, blocked** (no second working CAN board, no sender)	
C8	Testing and final report	**Not started** (no `RESULTS.md`; no 1-hour or overnight soak yet)	

4. Verified numbers

Host harness (real sources, ASan + UBSan): 434 checks flag off, 1493 TEST_STATS, 451 with the UART flag, 0 failed. Static checks: 58 pass on source, 61 with object code on the built images. Mutation: 141/144 caught (2026-11-09, not re-run on the C6 source changes); the U1 planted bugs: 5 of 5 caught. Board: C5 26 PASS / 0 FAIL / 1 skip; C6 idle caps 6 PASS; F24 flood runs at 0/3/8/24 PTA flooders.

5. Issues found and what happened

Issue	Status
O-01 FIQ overlaps the Core 1 stack	Resolved: FIQ stays masked (D1)
N-04 Linux did not reserve OP-TEE memory	**Resolved 2026-11-12** on the Pi; build rule not yet proven by a full image build
H1 hidden `memcpy` inside a lock hold	Fixed and re-verified
F24 any `/dev/tee0` user can delay PTA callers: worst frame delay 271 ms (3 flooders), 553 ms (8), 1.18 s (24); false traffic alarms, late answers	**Open, documented limit** (detection only). Unprivileged route closed on the Pi (`teeclnt` empty), not in the image build
F27 “events not rate-limited”	**Withdrawn**: the limiter exists (one per second per class); the ring wrapped because events stayed unread
Audit O-10 saturating miss counter	The code uses a plain increment (wrap needs about 136 years); recorded, not changed
My own mistakes	Wrong R/grace/N_req claim (they are defined in the original plan), two test bugs (T22, T17c), a relative build path that polluted the source tree, a false object-code “pass” on empty disassembly: all found, fixed and kept in the records

6. What is NOT done

UART never connected (W2, T23 and the UART line unverified on hardware); no CAN-bus evidence; Design 2 (chip B) not built; only 3 of 20+ boot IDs sampled; floods and gap histogram not run; IDLE_EXPIRED and the UART line not

flashed; 1-hour and overnight soaks not run; `RESULTS.md` not written.

7. Next

`monitor/PLAN_REMAINING_WORK_2026-11-12.md`: units U0-U8, decisions D-A to D-F, schedule to Saturday and the cut rule.